

to voting systems, intellectual property protection, and more, showcasing its versatility in fostering trust, transparency, and efficiency in diverse sectors.

A significant innovation in blockchain technology is the creation of self-executing agreements referred to as smart contracts, which are directly encoded into the system.

Smart contracts and decentralised applications (dApps)

Smart contracts are self-executing programs with predefined rules and conditions, deployed on a blockchain. These contracts automate and enforce the terms of agreements, cutting out the need for intermediaries and reducing the risk of errors or fraud. Decentralised applications, or dApps, are software applications that operate on a decentralised network of computers. Smart contracts are integral to the functionality of many dApps, providing a trustless and efficient way to execute code securely.

Decentralised applications leverage the decentralised nature of blockchain to provide users with transparent, secure, and censorship-resistant experiences. These applications operate on a peer-to-

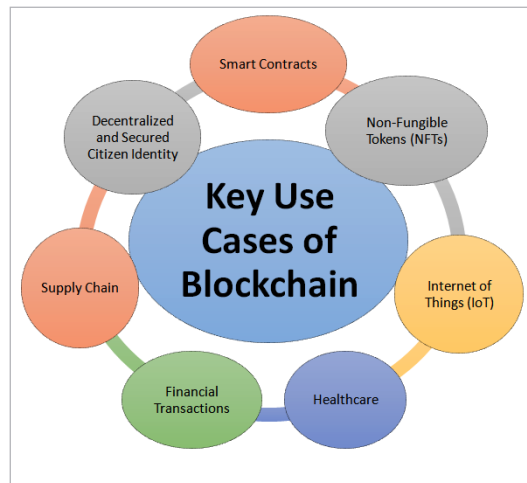


Figure 1: Service mesh architecture

peer network, ensuring that data remains tamper-proof and resilient to single points of failure. As the adoption of smart contracts and dApps continues to grow, their transformative impact on traditional business models becomes increasingly evident, ushering in a new era of decentralised and trustless interactions.

Programming platforms for the development of smart contracts and dApps

The development of smart contracts and dApps relies heavily on sophisticated programming platforms. These

platforms serve as the foundation for crafting secure, efficient, and decentralised solutions.

As the blockchain ecosystem expands, a diverse array of programming platforms has surfaced, each contributing to the evolution of smart contract and dApp development methodologies.

Ethereum

URL: ethereum.org

Ethereum is a pioneer in smart contract platforms and remains a dominant force in the space. Developers can use the Ethereum Virtual Machine (EVM) to deploy smart contracts written in languages like Solidity. Born out of the vision of Vitalik Buterin, Ethereum has introduced a versatile blockchain that extends beyond the limitations of Bitcoin, allowing developers to create complex dApps and execute smart contracts. Ethereum's blockchain serves as a decentralised and secure global computer, enabling a myriad of use cases across various industries.

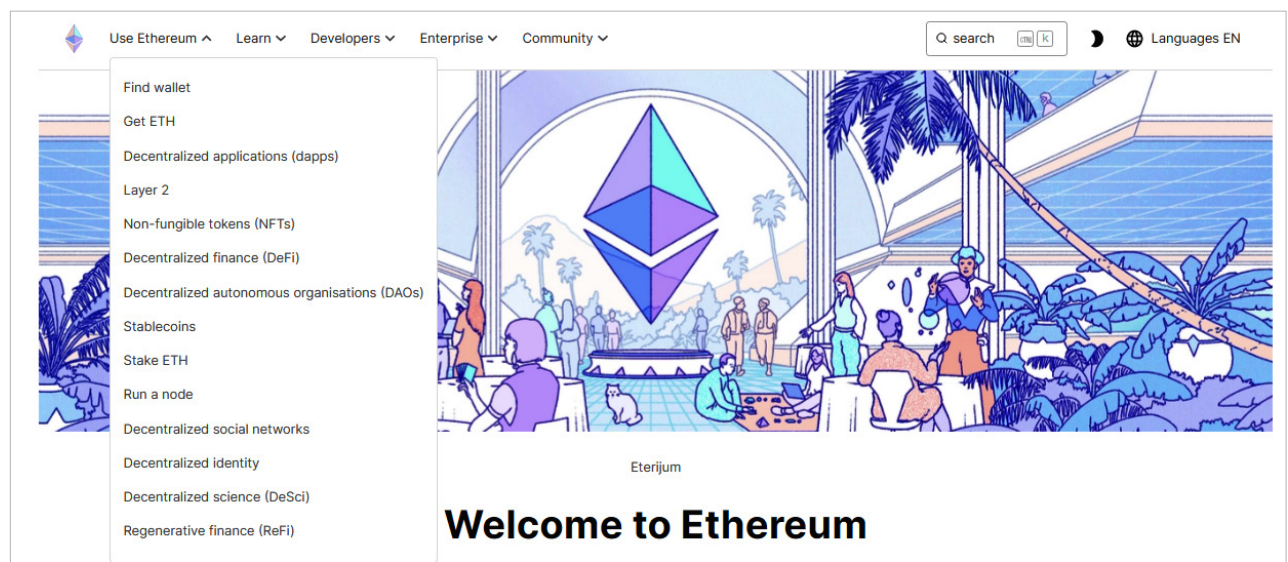


Figure 2: Ethereum blockchain